Outline Draft

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1 Introduction

Alzheimer's Disease (AD) is a complex neurodegenerative disease affecting more than 55 million people globally [14]. Symptoms are caused by the degeneration of the central nervous system, and are manifested by progressive memory loss, difficulty with language, drastic mood swings, among other effects. It is a disease that causes an extensive burden to those suffering from the disease through disorientation, to relatives in the form of mental taxation, as well as economically, projecting an estimated global economic burden of 16.9 trillion dollars by 2050 [18]. These factors are just a few reasons as to the importance of research in this area, where many scientists have worked to understand the underlying mechanisms of AD. However, much is still unknown in how AD is caused, as well as what biological processes occur in the brain, making it difficult to come to a decisive conclusion regarding its cause.

Many studies have been conducted in regard to understanding Alzheimer's Disease. With this, there are a few main hypotheses that are generally considered to be factors in the causation and progression of AD [22]. The *Amyloid Cascade Hypothesis* is among the most extensively studied in the literature, where it is proposed that a misfolding of β-amyloid (A_β) protein forms senile plaques. These plaques are hypothesized to be the main driving force in the intracellular deposition of misfolded τ proteins in neurofibrillary tangles (NFTs) and memory loss in the progression of Alzheimer's Disease [9]. Growing evidence, however, shows that this hypothesis is suggested to be but a small factor in the larger picture of the underlying operative mechanisms of the disease [12]. This has led researchers to branch into other pathways concerning AD, looking at the interactions between A_β and τ in an attempt to further understand the pathogenesis of these two factors [21]. Some studies have also observed the role of calcium in this process, further suggesting a more complex interaction that is occurring biologically. This occurs through a feedback loop between an upregulation of Ca^{2+} and the amyloidogenic pathway, remodeling the Ca^{2+} signaling pathway [4].

The various complex interactions in AD have shown to be large obstacles when modeling the behavior of the disease. Numerous different studies have in turn made an attempt at creating mathematical models observing different aspects of the disease in order to analyze the underlying dynamics. This has led to many models being introduced, each aiming to understand various components in Alzheimer's Disease through different methodologies. Compartments concerning the interaction between A_β and τ are studied, observing the suspected effects on neurodegeneration and cognitive decline through a mechanistic causal model [20, 8]. The feedback loop between A_β and Ca^{2+} has also been analyzed using a deterministic system, adding a noise-inducing element to the system [19, 23]. Models that observe lower-level processes have also been studied, lending to valuable insight as to what is seen on the cellular level [15, 13].

It should be noted that in these models, while each has its own way of observing some of the underlying mechanisms, there are still potential shortfalls when it comes to finding effective treatments. Evidence suggests that focusing solely on anti-A_β treatments is ineffective [1, 21, 24]. This provides difficulty in understanding exactly the importance of which component of the disease should be the focus of further study in order to provide evidence for treatments having higher efficacy. In this study, a mechanistic model is proposed, analyzing the interaction between A_β, Ca^{2+}, and τ proteins. We will observe the effects of this multifaceted interaction on neurodegeneration and cognitive decline. Sensitivity analysis is performed in order to better understand the importance of each aspect of the model, and optimal control is used in order to determine the suggested avenues of effective treatment.

2 Proposed Model

To understand the role that calcium may have in the development of AD, a model is proposed using three biomarkers found to be substantial during disease progression: beta-amyloid and tau, as well as the interactional biomarker calcium, which is suggested to have a role in beta-amyloid and tau protein accumulation. Also observed are neuronal loss and cognitive decline. The pathology of AD is suggested to start through a feedback loop between beta-amyloid and calcium through a dysregulation of calcium, which then exacerbates the aggregation of beta-amyloid, forming plaques. These plaques then promote the hyperphosphorylation of tau proteins, leading to neurodegeneration and subsequently cognitive decline. The model will describe the progression of Alzheimer's disease from a certain age; we use the arbitrary starting age of 50 in our model.

2.1 Amyloid Beta and Calcium Equations

Because amyloid beta and calcium are suggested to interact in a way that a dysregulation of calcium causes an acceleration of amyloid beta accumulation [19], this is modeled mathematically as

$$\frac{dA_\beta}{dt} = a_1 + a_2\frac{Ca}{Ca+\sigma} - k_1 A_\beta - u_1 A_\beta \quad \text{with} \quad A_\beta(T_0) = A_0, \tag{1}$$

where A_0 is the initial condition of A_β at age T_0, and can vary on a per patient basis [8]. The rate of A_β growth increases at a linear rate a_1 and decreases proportionally to the population of A_β, as described by $k_1 A_\beta$. Here, $a_2\frac{Ca}{Ca+\sigma}$ acts as the interaction of calcium on A_β using a Hill equation of $n = 1$. The removal rate can be modified by choosing different values for u_1 which represents the effect of treatment targeting A_β. The term u_1 can either be constant or be a function of time.

Similarly, an A_β interaction is observed to have a role in dysregulation of calcium, and the equation for calcium is defined as

$$\frac{dCa}{dt} = b_1 + b_2 A_\beta - k_2 Ca - u_2 Ca \quad \text{with} \quad Ca(T_0) = Ca_0. \tag{2}$$

Here, calcium grows at a linear rate b_1, and the accumulation due to A_β is positive at a rate b_2. There is also a natural removal rate of calcium, k_2. Notice that there is also the option to administer treatment targeting calcium through the term u_2.

2.2 Tau Equation

Many studies have suggested that the pathology of tau is initiated by amyloid beta accumulation, and have modeled this process mathematically [8, 25]. However, it has also been suggested that calcium dysregulation has an effect inducing the production of tau proteins [7]. Thus, the tau equation is given by,

$$\frac{d\tau}{dt} = c_1 + c_2 A_\beta + c_3 Ca - k_3\tau - u_3\tau \quad \text{with} \quad \tau(T_0) = \tau_0. \tag{3}$$

It is noted here that only pathological tau is observed, where other studies have used compartments relating to non-amyloid dependent tau accumulation, or suspected non-Alzheimer pathology (SNAP) [8, 20]. This function is similar to Equation (1) and Equation (2) in that $\frac{d\tau}{dt}$ grows at a linear rate d_1 and decreases proportional to the population of τ through $k_3\tau$. There is also the treatment term $u_3\tau$.

2.3 Neurodegeneration Equation

When tau is in a pathological state, it no longer binds to microtubules as intended; instead, the formation of NFTs occur in the neurons, leading to neurodegeneration [2]. Hence, this equation is defined as

$$\frac{dN}{dt} = d_1 + d_2\tau - k_4 N.$$

This equation follows the established structure and also includes the interaction term $d_2\tau$. There are no treatment terms for the N and C equations.

2.4 Cognitive Decline Equation

Both tau and neurodegeneration are shown to directly affect cognitive decline [3, 8]. Because of this, the equation for cognitive decline is defined as

$$\frac{dC}{dt} = e_1 + e_2 NR + e_3\tau - k_5 C.$$

The parameter R represents a patient having the APOE $\epsilon 4$ gene, which has been shown to be a risk factor in Alzheimer's disease [20].

This model is summarized as the following system of ordinary differential equations:

$$\begin{aligned}
\frac{dA_\beta}{dt} &= a_1 + a_2\frac{Ca}{Ca+\sigma} - k_1 A_\beta - u_1 A_\beta \\
\frac{dCa}{dt} &= b_1 + b_2 A_\beta - k_2 Ca - u_2 Ca \\
\frac{d\tau}{dt} &= c_1 + c_2 A_\beta + c_3 Ca - k_3\tau - u_3\tau \\
\frac{dN}{dt} &= d_1 + d_2\tau - k_4 N \\
\frac{dC}{dt} &= e_1 + e_2 NR + e_3\tau - k_5 C
\end{aligned} \tag{4}$$

with the initial conditions:

$$\begin{cases} A_\beta(T_0) = A_0 \\ Ca(T_0) = Ca_0 \\ \tau(T_0) = \tau_0 \\ N(T_0) = N_0 \\ C(T_0) = C_0. \end{cases}$$

3 Analysis

3.1 Preliminary Results

Assuming system (4) is subject to non-negative initial conditions, the system is invariant in the non-negative orthant. Let $r_n = k_n + u_n$ for the simplicity of calculations. Considering the components uncoupled from the rest of the proposed model, dA_β/dt and dCa/dt can be shown to be bounded. First, notice that

$$a_1 + a_2 \frac{Ca}{Ca + \sigma} - r_1 A_\beta \leq a_1 + a_2 - r_1 A_\beta$$

since $Ca/(Ca + \sigma) \in [0, 1)$. So, $a_1 + a_2 - r_1 A_\beta < 0 \implies dA_\beta/dt < 0$ when $A_\beta > \frac{a_1+a_2}{r_1}$. Similarly, using the fact A_β is bounded, it follows that

$$\frac{dCa}{dt} = b_1 + b_2 A_\beta - r_2 Ca \leq b_1 + b_2 M - r_2 Ca$$

as $M > A_\beta(t)$ for all $t \geq 0$. So, $b_1 + b_2 M - r_2 Ca < 0 \implies dCa/dt < 0$ when $Ca > \frac{b_1+b_2 M}{r_2}$, thus Ca is also bounded. The rest of the system follows from these two variables, so the rest of the system is bounded as well.

Lemma 1 (Establishment of Unique Equilibria of Uncoupled System)**.** *There exists a unique equilibrium (A_β^*, Ca^*) for the system (A_β, Ca).*

Proof. To establish the existence of equilibria for the system, a solution to the following system of equations must be obtained:

$$\begin{cases} \frac{dA_\beta}{dt} = a_1 + a_2 \frac{Ca}{Ca+\sigma} - r_1 A_\beta = 0 \\ \frac{dCa}{dt} = b_1 + b_2 A_\beta - r_2 Ca = 0. \end{cases}$$

By solving dA_β/dt for A_β, we get

$$A_\beta = \frac{a_1 Ca + a_2 Ca + a_1 \sigma}{r_1(Ca + \sigma)}.$$

We then substitute this value into dCa/dt. The resulting expression is

$$b_1 - r_2 Ca + \frac{b_2(a_1 Ca + a_2 Ca + a_1 \sigma)}{r_1(Ca + \sigma)}. \tag{5}$$

Setting Equation (5) equal to 0 and doing further simplification yields the quadratic,

$$-(r_1r_2)Ca^2 + (a_1b_2 + a_2b_2 + b_1r_1 - r_1r_2\sigma)Ca + a_1b_2\sigma + b_1r_1\sigma = 0. \tag{6}$$

Notice that in Equation (6) the quadratic term is always negative and the constant term is always positive. The linear term can either be positive or negative depending on the choice of parameters. Regardless, there is only one sign change in the expression, so by Descartes' Rule of Signs, there is one positive real root to this quadratic. Thus, there exists a unique positive equilibrium (A_β^*, Ca^*) for the submodel. □

Lemma 2 (Global Stability of Equilibrium). *The equilibrium (A_β^*, Ca^*) is globally asymptotically stable.*

Proof. Let $\mathbf{F}(A_\beta, Ca) = \left\langle a_1 + \frac{a_2Ca}{Ca+\sigma} - r_1A_\beta,\, b_1 + b_2A_\beta - r_2Ca\right\rangle$. Then, $\nabla \cdot \mathbf{F} = -r_1 - r_2$. Choose the function $\alpha(A_\beta, Ca) = 1$. So, $\nabla \cdot (\alpha\mathbf{F}) = -r_1 - r_2 < 0$, therefore by the Bendixson-Dulac Theorem, there is no limit cycle in the system of ODEs. So, the equilibrium point in the non-negative orthant must be globally asymptotically stable. □

Theorem 1 (Global Stability of Equilibrium of System). *There exists a unique globally asymptotically stable equilibrium $(A_\beta^*, Ca^*, \tau^*, N^*, C^*)$.*

Proof. By Lemma 2, there is a unique globally asymptotically stable equilibrium (A_β^*, Ca^*) for the (A_β, Ca) submodel. By solving (3) for τ we get the following equilibrium value:

$$\tau^* = \frac{c_1 + c_2A_\beta^* + c_3Ca^*}{k_3 + u_3}.$$

By similar methods we find

$$N^* = \frac{d_1 + d_2\tau^*}{k_4}$$

and

$$C^* = \frac{e_1 + e_2N^*R + e_3\tau^*}{k_5}.$$

Thus there is a unique global asymptotically stable equilibrium solution for system (4). □

4 Parameters

The values for the 19 key parameters in the causal model are obtained and estimated from the literature, and are summarized in Table 1. In order to provide a plausible model, parameters used in previous studies are adjusted to allow for similar units and orders of magnitude. When this is not possible, parameters are estimated using values found from various biological studies as are credible.

Table 1: Parameters used for simulation

Var	Value	Unit	Source
A_0	0	micromolars (μM)	-
Ca_0	100	nanomolars (nM)	[16]
τ_0	0	μM	-
N_0	0	billions of neurons lost	-
C_0	0	MMSE test score decline	-
a_1	6.5641	μM $\cdot$ year^{-1}	[15] adjusted
a_2	1.5778	μM $\cdot$ year^{-1}	[23] adjusted
b_1	315569260	nM $\cdot$ year^{-1}	[23] lets review this one
b_2	6311385.2	year^{-1} (100)	[23]
c_1	52.2958	μM $\cdot$ year^{-1}	[17]
c_2	1.78367	year^{-1}	[15]
c_3	0.1	year^{-1} (1/100)	no source
d_1	0.0716	billions of neurons lost $\cdot$ year^{-1}	no source
d_2	0.398406	billions of neurons lost $\cdot$ year^{-1} $\cdot$ μM^{-1}	[17]
e_1	0.01463	MMSE decline $\cdot$ year^{-1}	no source
e_2	0.008684	MMSE decline $\cdot$ year^{-1} $\cdot$ billions neurons lost^{-1}	[8] adjusted
e_3	0.01992	MMSE decline $\cdot$ year^{-1} $\cdot$ μM^{-1}	[8] adjusted
k_1	0.303598	year^{-1}	[15] adjusted
k_2	3155692.6	year^{-1}	[23]
k_3	1.8333	year^{-1}	no source
k_4	0.3588	year^{-1}	no source (?)
k_5	0.08684	year^{-1}	no source
u_1	varies	year^{-1}	-
u_2	varies	year^{-1}	-
u_3	varies	year^{-1}	-
σ	$50 - 150$	nM	no source (?)

5 Sensitivity Analysis

Sobol sensitivity analysis is used to provide a quantitative assessment of the impact that model parameters have on the causal dynamics of the proposed model. This computation was performed using Python with the SALib library [10, 11]. The first, second, and total-order indices (S_T) were computed using different sample sizes, ranging from $N = 2^4$ to $N = 2^{16}$. Using the parameter ranges from Table 1, the first-order and total-order sensitivity indices are highlighted in Figure 1; initial conditions were set for $Ca(0) = 100$, and zero for the rest of the state variables. In order to validate these indices, Figure 2 shows the six most sensitive parameters for each sample size. The total-order indices are quantified along with the confidence interval for each parameter at $N = 2^{16}$ in Table 2. Using a threshold of 10^{-4}, the parameters falling below this threshold provide a neglible impact on the overall model. From this, it is observed that the growth of calcium in Equation (3) is suggested to have very little effect on tau protein synthesis, which coroborates with findings in some studies (list citations).

Table 2: Total Sobol Sensitivity indices for various parameters with corresponding confidence intervals for $N = 2^{16}$

Param	k5	k3	e3	k1	c1	c2	a1
ST	2.499e-01	2.871e-01	1.283e-01	6.081e-02	5.787e-02	4.916e-02	3.960e-02
Conf Int	3.382e-03	3.768e-03	1.803e-03	8.049e-04	7.786e-04	6.181e-04	4.895e-04

Param	k4	e2	r	d2	k2	c3	a2
ST	3.111e-02	3.103e-02	3.100e-02	3.087e-02	5.261e-03	5.140e-03	2.289e-03
Conf Int	3.951e-04	4.445e-04	4.254e-04	4.166e-04	7.286e-05	7.242e-05	3.007e-05

Param	b1	b2	e1	d1	sig
ST	2.180e-03	6.361e-04	1.691e-05	3.243e-07	3.926e-16
Conf Int	3.303e-05	1.013e-05	2.063e-07	7.372e-09	5.648e-16

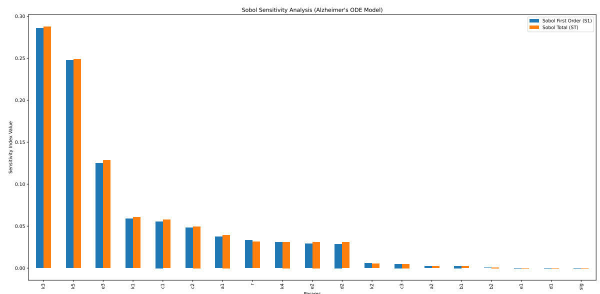

Figure 1: First and Total order indices at $N = 2^{10}$

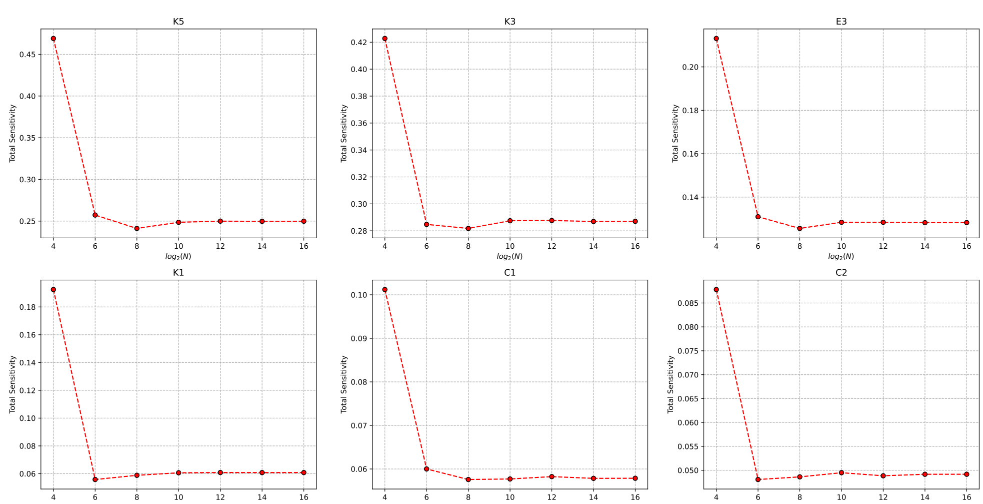

Figure 2: 6 most sensitive parameters for $N \in [2^4, 2^{16}]$

6 Results

Using the derived biological parameters in Table 1, we run the simulation forward for 50 years with the initial conditions $(A_0, Ca_0, \tau_0, N_0, C_0) = (0, 100, 0, 0, 0)$. This simulation uses solve_ivp from the SciPy package in Python, and the results from the simulation are shown in Figure 3. Initially, A_β and Ca increase and drive each other's growth. This leads to a large growth of τ proteins, thus increasing neuron loss over time. The biomarker populations reach their equilibrium values relatively early, but it takes cognitive decline many years to reach its equilibrium value.

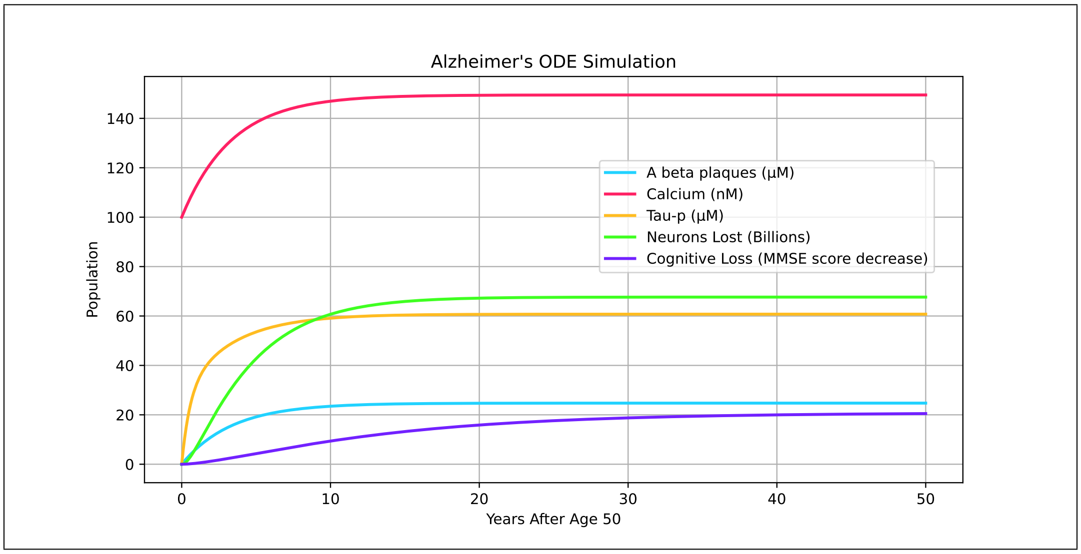

Figure 3: ODE system simulation

A phase portrait, shown in Figure 4, displays how different trajectories with varying initial conditions traverse the $A_\beta - Ca$ space. Trajectories will quickly move toward the Ca nullcline with slight movement along the horizontal axis. This is because the time-scale of Ca is much smaller than that of A_β. Once trajectories are nearby the nullcline, they along the nullcline toward the equilibrium solution.

To determine which proteins are most effective to treat, we graph the equilibrium cognitive decline value against varying treatments against each protein, as seen in Figure 5. Each u_n term increases such that the behavior of the function can be observed. We see that the variable that has the largest impact on C^* is u_3, the treatment that reduces the number of τ proteins. The next most effective treatment is u_1, which reduces the concentration of A_β plaques in the brain. We observe that treating the average level of Ca does not have a large effect on the equilibrium cognitive decline.

7 Optimal Control

We use optimal control to determine the optimal values of u_n to minimize the total cost of treatments and results. In optimal control, we will let each u_n be a function of time, $u_n(t)$, as it is not realistic to maintain a constant treatment long-term. Tracking the cost

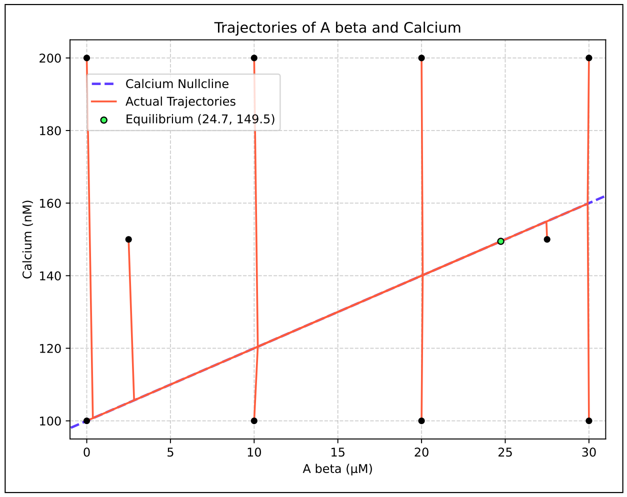

Figure 4: Phase portrait

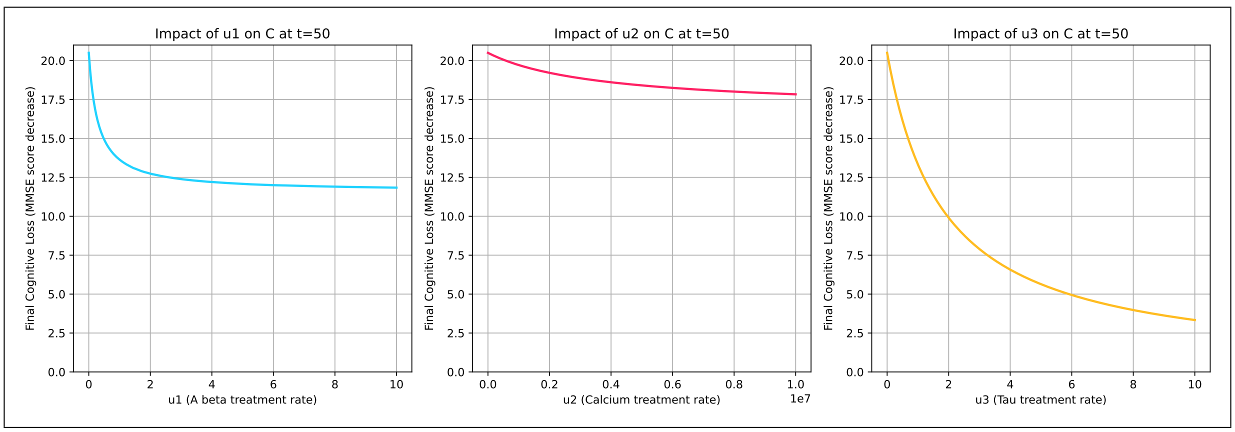

Figure 5: Treatment effect on C^*

of treatments is useful for our application as some drugs used to treat Alzheimer's Disease have been shown to induce strong side effects such as brain swelling and brain bleeding [5]. So, minimizing this cost function while maximizing treatments will let the medication have its strongest impact without adverse effects.

7.1 Objective Function

The following equation is our objective function which tracks our cost over the time horizon of the treatment.

$$J(u_1, u_2, u_3) = \int_0^K A_1 N + A_2 C + A_3 u_1^2 + A_4 u_2^2 + A_5 u_3^2 \, dt. \tag{7}$$

The goal of treatment is to minimize the equilibrium values N^* and C^* while at the same time minimizing the side effects from the drugs. The terms A_n are the weights which determine how important each associated variable is in the optimization problem. A_3, A_4, and A_5 are the weights associated with the quadratic cost of treatments u_n that target A_β, calcium, and τ protein levels. The set of admissible control functions is given by:

$$u = \left\{(u_1, u_2, u_3) \in L^\infty(0, K)^3 \mid (u_1, u_2, u_3) \in [0, u_1^{\max}], [0, u_2^{\max}], [0, u_3^{\max}] \; \forall\, t \in [0, K]\right\}$$

7.2 Hamiltonian

$$\begin{aligned}\mathcal{H} = A_1 N + A_2 C + A_3 u_1^2 + A_4 u_2^2 + A_5 u_3^2 + \lambda_1 \left(a_1 + a_2 \frac{Ca}{Ca+\sigma} - k_1 A_\beta - u_1 A_\beta\right)\\ + \lambda_2 \left(b_1 + b_2 A_\beta - k_2 Ca - u_2 Ca\right) + \lambda_3 \left(c_1 + c_2 A_\beta + c_3 Ca - k_3 \tau - u_3 \tau\right)\\ + \lambda_4 \left(d_1 + d_2 \tau - k_4 N\right) + \lambda_5 \left(e_1 + e_2 N R + e_3 \tau - k_5 C\right).\end{aligned}$$

The Hamiltonian includes the treatment costs as described in Equation (7) as well as the co-state vectors for each variable present in our system. Thus, we seek an optimal control set (u_1^*, u_2^*, u_3^*) such that

$$J(u_1^*, u_2^*, u_3^*) = \min\left\{J(u_1, u_2, u_3) \mid (u_1, u_2, u_3) \in \mathcal{U}\right\}.$$

Theorem 2 (Existence of Optimal Control)**.** *There exists an optimal control set (u_1^*, u_2^*, u_3^*) that minimizes $J(u_1, u_2, u_3)$ over $\mathcal{U}$. Furthermore, there exists adjoint variables $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ and λ_5 that satisfy the following system of differential equations:*

$$\begin{aligned}\frac{d\lambda_1}{dt} &= -\frac{\partial \mathcal{H}}{\partial A_\beta} = k_1\lambda_1 + u_1\lambda_1 - b_2\lambda_2 - c_2\lambda_3\\ \frac{d\lambda_2}{dt} &= -\frac{\partial \mathcal{H}}{\partial Ca} = -\lambda_1 \frac{a_2\sigma}{(Ca+\sigma)^2} + k_2\lambda_2 + u_2\lambda_2 - c_3\lambda_3\\ \frac{d\lambda_3}{dt} &= -\frac{\partial \mathcal{H}}{\partial \tau} = k_3\lambda_3 + u_3\lambda_3 - d_2\lambda_4 - e_3\lambda_5\\ \frac{d\lambda_4}{dt} &= -\frac{\partial \mathcal{H}}{\partial N} = -A_1 + k_4\lambda_4 - e_2 R\lambda_5\\ \frac{d\lambda_5}{dt} &= -\frac{\partial \mathcal{H}}{\partial C} = -A_2 + k_5\lambda_5\end{aligned} \tag{8}$$

$$\begin{aligned}\frac{d\lambda_1}{dt} &= -\frac{\partial \mathcal{H}}{\partial A_\beta} = (k_1 + u_1)\lambda_1 - b_2\lambda_2 - c_2\lambda_3\\ \frac{d\lambda_2}{dt} &= -\frac{\partial \mathcal{H}}{\partial Ca} = -\frac{a_2\sigma_1\lambda_1}{(Ca+\sigma_1)^2} + (k_2 + u_2)\lambda_2 - \frac{c_3\sigma_2\lambda_3}{(Ca+\sigma_2)^2}\\ \frac{d\lambda_3}{dt} &= -\frac{\partial \mathcal{H}}{\partial \tau} = (k_3 + u_3)\lambda_3 - d_2\lambda_4 - e_3\lambda_5\\ \frac{d\lambda_4}{dt} &= -\frac{\partial \mathcal{H}}{\partial N} = -A_2 + k_4\lambda_4 - e_2 R\lambda_5\\ \frac{d\lambda_5}{dt} &= -\frac{\partial \mathcal{H}}{\partial C} = -A_1 + k_5\lambda_5\end{aligned} \tag{9}$$

with transversality conditions $\lambda_1(K) = \lambda_2(K) = \lambda_3(K) = \lambda_4(K) = \lambda_5(K) = 0$. Also, the optimal control set has the characterization:

$$u_1^* = \min\left\{u_1^{\max}, \max\left\{0, \frac{\lambda_1 A_\beta}{2A_3}\right\}\right\}$$
$$u_2^* = \min\left\{u_2^{\max}, \max\left\{0, \frac{\lambda_2 Ca}{2A_4}\right\}\right\}$$
$$u_3^* = \min\left\{u_3^{\max}, \max\left\{0, \frac{\lambda_3 \tau}{2A_5}\right\}\right\}.$$

Proof. Using Corollary 4.1 of [6], the existence of an optimal control set follows from the Lipschitz property of system (4) with respect to the state variables, *a priori* boundedness of the state solutions established in Subsection 3.1, and the convexity of the integrand of J with respect to (u_1, u_2, u_3).

By applying Pontryagin's Maximum Principle, the adjoint variables must satisfy system (8) with zero final time conditions. To obtain the characterizations of the optimal controls, we solve $\frac{\partial \mathcal{H}}{\partial u_1} = \frac{\partial \mathcal{H}}{\partial u_2} = \frac{\partial \mathcal{H}}{\partial u_3} = 0$ on the interior of the control set $\mathcal{U}$. Using the bounds on the controls, we obtain the desired characterization. □

7.3 Numerical Results

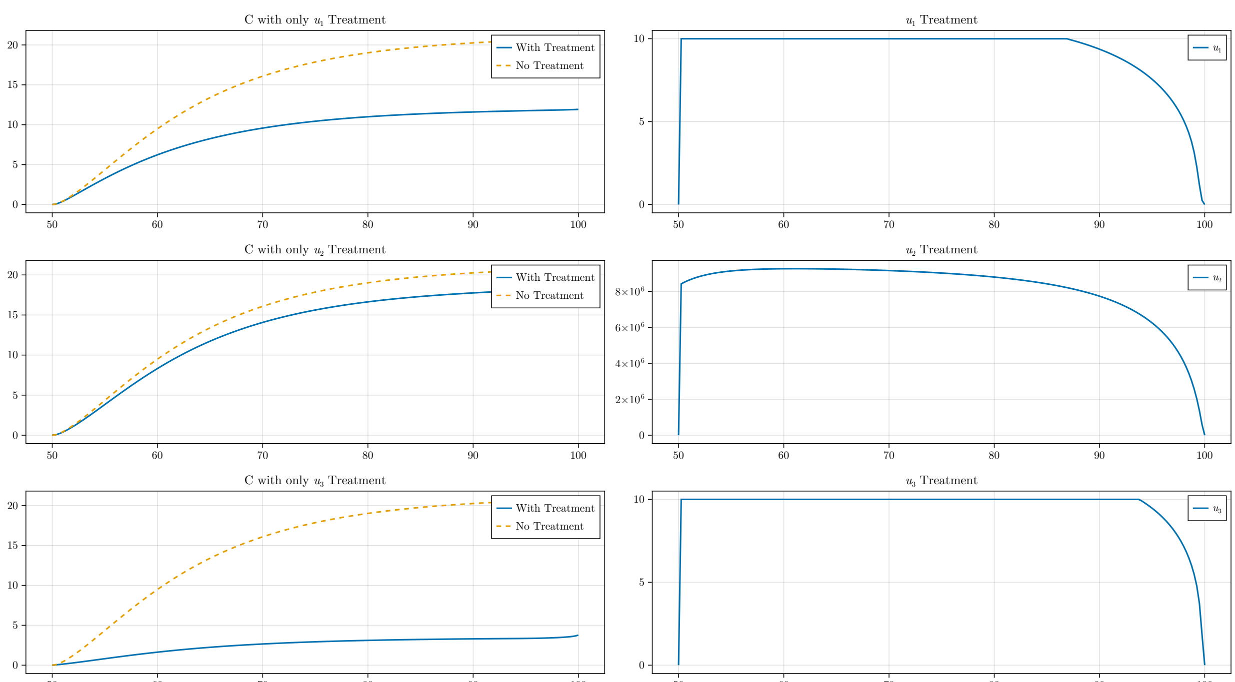

Figure 6: Optimal control simulation results

- explain how weights were calculated

Figure 6 shows the values of A_β, calcium, and tau proteins over time with and without treatments, as well as the value of u_n over the duration of the treatment. After running numerical simulations for optimal control, we find that the results are in line with the simulation shown in Figure 5. That is, treatments that target tau protein are most effective in reducing C^*, followed by treatments targeting A_β. Again, treatments targeting Ca are found to be the least effective.

8 Conclusion

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